

CO 450/650: Combinatorial Optimization

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Contents

1	Minimum Cost Spanning Trees	4
1.1	Minimum Spanning Tree Problem	4
1.2	Kruskal's Algorithm	6
1.3	Correctness via LP	7
1.3.1	Integer Programming Formulation	7
1.3.2	LP relaxation:	8
1.3.3	Greedy and Max Cost Forest	9
1.3.4	Kruskal for Maximum Cost Forest	10
2	Matroids	11
2.1	Matroid 1	11
2.2	Matroid 2	13
2.3	Matroid 3	16
2.3.1	Circuit characterization	16
2.4	Matroid 4	18
2.5	Polymatroids	18
2.6	Matroid Construction	21
3	Matchings	24
3.1	Matchings 1	24
3.2	Matching 2	26
3.3	Matching 3	28
3.4	Matching 4	30
3.5	Matching 5	32
3.6	Matching 6	35
4	Weighted Matching	36
4.1	Weighted Matching 1	36

1 Minimum Cost Spanning Trees

What's a spanning tree?

Definition 1.1

Given a graph $G = (V, E)$, a subgraph T is a spanning tree of G if:

- $V(T) = V(G)$
- T is connected
- T is acyclic (contains no cycle)

Theorem 1.2

Let $G = (V, E)$ be connected graph, T be a subgraph of G , with $V(T) = V$, then the following are equivalent (TFAE)

- T is a spanning tree of G .
- T is minimally connected (T will be disconnected if any edge is dropped).
- T is maximally acyclic (Add any edge between vertices of T makes it cyclic).
- $\forall u, v \in V$, there exists a unique $u - v$ path in T (call it $T_{u,v}$).

Theorem 1.3

A graph $G = (V, E)$ is connected if and only if $\forall A \subseteq V$ with $\emptyset \neq A \neq V$, we have $\delta(A) \neq \emptyset$ ($\delta(A) := \{e \in E : |e \cap A| = 1\}$, the set of edges with exactly one edge in A).

1.1 Minimum Spanning Tree Problem

Input:

- Connected graph $G = (V, E)$.
- Costs $C_e, \forall e \in E$.

Output: A spanning tree T of G of minimum cost $C(T) := \sum_{e \in E(T)} C_e$.

Theorem 1.4

Let $G = (V, E)$, connected, $C : E \mapsto \mathbb{R}$, T is a spanning tree of G , then TFAE:

- a) T is a MST (minimum spanning tree).
- b) $\forall uv \in E \setminus E(T)$, all edges e on $T_{u,v}$ have $C_e \leq C_{uv}$.
- c) $\forall e \in E(T)$, let T_1, T_2 be the two connected components obtained from T when removing e . then e is a min cost edge in $\delta(T_1)$ (of G).

Proof.

- a) $\implies$ b). Suppose $\exists uv \in E \setminus E(T)$ and $e \in T_{u,v}$ such that $C_e > C_{uv}$, consider $T' = T + uv - e$. Since we don't change delete any vertices, $V(T') = V(T) = V(G)$. If we write $T_{u,v} = u, v_1, \dots, v_n, v$ and say v_i, v_{i+1} are the ends of e . Then, for any two vertices of G , if they are not on $T_{u,v}$, they are still connected. If at least one of them is on $T_{u,v}$, say k_2 is on $T_{u,v}$, WLOG, say the unique path is $k_1, \dots, u, \dots, k_2$, if e is not in the path, we are good, if it is, then we take $k_1, \dots, u, v, \dots, k_2$ on T' . Hence, T' is connected. That is, T' is connected and $|E(T')| = |E(T)| = |V| - 1$, so by theorem 2, we know T' is a spanning tree of G . And since $C_{uv} < C_e$, $C(T') < C(T)$, T is not a MST, contradiction, so no such uv exists.
- b) $\implies$ c). Suppose $\exists e \in T, uv \in \delta(T_1)$ such that $C_{uv} < C_e$. First, $uv \notin E(T)$, otherwise, since $v \in T_2$ and T_1, T_2 are connected, so there is a cycle including uv and e in T , contradiction. Also, $e \in T_{uv}$, because we have $u \in T_1$, and $v \in T_2$, and T_1, T_2 are separated by e , so any path from T_1 to T_2 will include e . Then this contradicts to b), contradiction, no such e exists, c) is true.
- c) $\implies$ a). Let T satisfy c). Let T^* be a MST with largest $k := |E(T) \cap E(T^*)|$. If $k = n - 1 = |V| - 1$, we are done. Else, there is $e \in E(T) \setminus E(T^*)$ (note T is also a spanning tree). Let T_1, T_2 be connected component of $T - e$, there exists $e^* \in E(T^*) \cap \delta(T_1)$. First $e^* \notin E(T)$ because otherwise, we have e and e^* connecting T_1 and T_2 in T (note $e \neq e^*$). Also, $T' = T^* - e^* + e$ is also a spanning tree, because all vertices stay connected and the number of edges stay the same (as above proof). By c), $C_e \leq C_{e^*}$, so $C(T) \leq C(T^*)$. So T' is a MST, and $|E(T) \cap E(T')| = k + 1 > k$, contradiction. So $k = n - 1$, $T = T^*$ which is a MST.

□

1.2 Kruskal's Algorithm

Algorithm 1 Kruskal's Algorithm for MST

Input: G be a connected graph, $n = |V|$, $m = |E|$
 $H = (V, \emptyset)$
while H is not a spanning tree **do**
 Find the cheapest edge whose endpoints are in different components of H
 $H \leftarrow H + e$
end while

We also have an equivalent version:

Algorithm 2 Equivalent

Sort edges so that $C_{e_1} \leq \dots \leq C_{e_m}$.
for $i = 1, \dots, m$ **do**
 if endpoints u, v of e_i are in different components of H **then**
 $H \leftarrow H + e_i$
 end if
end for

Implementation:

- Keep array *comp*, with $\text{comp}[v] \leftarrow v, \forall v \in V$ initially.
- The if step in algorithm 2 can be done by checking if $\text{comp}[u] == \text{comp}[v]$, for $e = uv$. $O(1)$.
- When the assignment step in alg2 is executed, go through $\text{comp}[t], \forall t \in V$, if $\text{comp}[t] == \text{comp}[v]$, set $\text{comp}[t] = \text{comp}[u]$. That is, make sure u, v and the vertices they are connected to are in the same component. $O(n)$.
- Sort step $O(m \log m)$.
- For loop step $O(m)$ in total.

Overall, we have $O(m \log m) + O(mn) = O(mn)$, which is a polynomial time. At the end, H will be a spanning tree.

Q: Can alg1 get stuck?

- The e we need to find always exists. Since H is not a spanning tree, either it is not connected, or it has a cycle in it. However, if H has a cycle, the last edge added to that cycle will not be added because its two endpoints are already in H . Hence, H is disconnected, so we can find an edge connecting different components of H .
- Everytime $H \leftarrow H + e$ is executed, two different components are connected, so the number of different connected components of H minus 1. Also, since H is acyclic before the assignment, and e connects two different components, there is no cycle.

- Every iteration, the number of components minus 1, and we have n components at the beginning, so we do $O(n)$ iterations, during the time, we keep the H acyclic.

Q: Does it return a MST?

Suppose not, there exists $uv \in E \setminus E(H)$ and $e \in H_{uv}$ with $C_{uv} < C_e$ by theorem4, so C_{uv} will be considered before C_e in the first step of alg2. When C_{uv} is being tested in alg2's third step, there is no path from u to v , so they are in different components, so uv will be added to H , not e , contradiction.

1.3 Correctness via LP

- Show techniques that can be used in other settings
- Lead to "good" approaches for more challenging problems

1.3.1 Integer Programming Formulation

- Let $x_e \in \{0, 1\}$ to indicate if edge e is in the MST
- Spanning tree: acyclic, $n - 1$ edges ($n := |V|$)

$$\begin{aligned}
 \min \quad & \sum_{e \in E} c_e x_e \quad c^T x \\
 s.t. \quad & x(E) = n - 1, \text{ where } x(F) := \sum_{e \in F} x_e \\
 & x(F) \leq n - \kappa(F), \forall F \subseteq E \\
 & x \in \{0, 1\}^E
 \end{aligned}$$

For **Acyclic**:

Consider $F \subseteq E$. How many edges of F can a spanning tree have?

Let $\kappa(F)$ be the number of connected components of (V, F) , then our answer is $n - \kappa(F)$. Since if we consider every connected components of (V, F) , we can find a spanning tree in it and have at most the number of vertices in that component minus one edges. So, sum over all components, we have $n - \kappa(F)$ at most without forming a cycle.

Note: If $F = \{e\}$, then $\kappa(F) = n - 1$, so $x(F) \leq n - \kappa(F)$ becomes $x_e \leq 1$, so our problem becomes

$$\begin{aligned}
 \min \quad & \sum_{e \in E} c_e x_e \\
 s.t. \quad & x(E) = n - 1 \\
 & x(F) \leq n - \kappa(F), \forall F \subseteq E \\
 & x \geq 0, x \in \mathbb{Z}^E
 \end{aligned}$$

1.3.2 LP relaxation:

$$\begin{aligned}
 (P_{ST}), \zeta_{P_{ST}}^* &:= \min \sum_{e \in E} c_e x_e \\
 \text{s.t. } &x(E) = n - 1 \\
 &x(F) \leq n - \kappa(F), \forall F \subseteq E \\
 &x \geq 0
 \end{aligned}$$

It has optimal solutions. Since G is connected, it has feasible solutions (just find a spanning tree) and its feasible regions is bounded, so it has optimal solutions.

Proof Idea:

- Any spanning tree T corresponds to a feasible solution to $(P_{ST}) \implies c(T) \geq \zeta_{P_{ST}}^*$.
- Shows that spanning tree produced by Kruskal is optimal for (P_{ST}) (using Complementary Slackness).

$$\begin{aligned}
 \min & c^T x \\
 \text{s.t. } &x(E) = n - 1 \\
 &x(F) \leq n - \kappa(F), \forall F \subseteq E \\
 &x \geq 0
 \end{aligned}$$

note that $n - 1 = n - \kappa(E)$. Then we find the dual

Dual (D_{ST}) :

$$\begin{aligned}
 \max & \sum_{F \subseteq E} (n - \kappa(F)) y_F \\
 \text{s.t. } & \sum_{F: e \in F} y_F \leq c_e, \forall e \in E \\
 & y_F \leq 0, \forall F \subset E \\
 & y_E \text{ is free.}
 \end{aligned}$$

Let $E = \{e_1, \dots, e_m\}$, with $c_{e_1} \leq c_{e_2} \leq \dots \leq c_{e_m}$. Let $E_i = \{e_1, \dots, e_i\}$,

- $\bar{y}_{E_i} = c_{e_i} - c_{e_{i+1}} \leq 0, \forall i = 1, \dots, m - 1$
- $\bar{y}_E = c_{e_m}, \bar{y}_F = 0, \forall \text{ other } F$.

Now, we want to show that $\bar{y}$ is feasible for (D_{ST}) and all constraints are satisfied at equality (except the $y_F \leq 0$ ones). For each $e_i \in E$, we know $e_i \in E_i, \dots, E_m$ and some other non- E_i edge subsets. Hence,

$$\begin{aligned}
 \sum_{F: e_i \in F} y_F &= \sum_{j=i}^m y_{E_j} + \sum_{F \neq E_j, j \geq i: e_i \in F} y_F \\
 &= c_{e_i} - c_{e_m} + c_{e_m} + 0 \\
 &= c_{e_i}
 \end{aligned}$$

So the Complementary Slackness condition for Dual constraints are satisfied, we only need to check either $\bar{y}_i = 0$ or $\bar{x}_{E_i}$ constraint is tight.

Now, let $\bar{x}$ be the incidence vector of tree T constructed by Kruskal. Note: $\bar{x}(E_i) = \sum_{e \in E_i} \bar{x}_e = |E(T) \cap E_i|$.

- $T_i = (V, E_i \cap E(T))$ is a maximally acyclic subgraph of $H_i = (V, E_i)$. Suppose not, then we can add an edge e_k of $E_i \setminus E(T)$ to T_i , and it's still acyclic. This edge e_k connects two component of T_i , otherwise, since $e_k \notin E(T)$, its endpoints are in the same component in T , so there is a path between its endpoints in $E_k \cap E(T) \subseteq E_i \cap E(T)$, contradiction. Since it connects two components of T_i , it will added be to T at k^{th} iteration, so it will be in $E(T)$, contradiction.
- As argued before, $n - \kappa(E_i)$ is the largest number of edges we can choose from E_i without forming a cycle in $H_i = (V, E_i)$, that is, by previous point, $n - \kappa(E_i) = |E_i \cap E(T)| = \bar{x}(E_i)$.
- Now we argue the Complementary Slackness conditions are satisfied. For each $F \subseteq E$, if $F \in \{E_1, \dots, E_m\}$, then by the previous point, the equality is tight; otherwise, $y_F = 0$. For each $e \in E$, we showed that all constraints of the dual problem are tight. So $\bar{x}, \bar{y}$ are optimal for P_{ST}, D_{ST} respectively.
- Hence, $c^T \bar{x} = c(T) = \zeta_{P_{ST}}^*$ by Complementary Slackness Theorem.

Consequence of Proof:

- $\zeta_{P_{ST}}^* = c(T^*)$, where T^* is MST.
- Solving the above LP can give us an integral solution (under mild assumptions), which rarely happens.

Alternative Formulation for P_{ST} :

$$\begin{aligned} \zeta_{P_{ST}}^* &:= \min c^T x \\ \text{s.t. } &x(E) = n - 1 \\ &x(E(S)) \leq |S| - 1, \forall \emptyset \subsetneq S \subsetneq V \\ &x \geq 0 \end{aligned}$$

where $E(S) = \{e \in E : |e \cap S| = 2\}$.

1.3.3 Greedy and Max Cost Forest

- MST Algorithms are greedy (best decision based only on local structure).
- Ex: Max weight independent set. Given $G = (V, E)$, $S \subseteq V$ is an independent set if $\forall u, v \in S, uv \notin E$. Then given $C_v, \forall v \in V$, find independent set S , which maximize $c(S) := \sum_{v \in S} C_v$.

Maximum Forest Problem:

Given $G = (V, E)$, a forest is a subgraph (V, F) with $F \subseteq E$ that is acyclic. (We refer to a forest by its set of edges). Then we want

Given $G = (V, E)$, $c_e, \forall e \in E$, find a forest F maximizing $c(F) := \sum_{e \in F} c_e$.

USE MST:

- Compute MST with respect to $c'_e = -c_e$.
- Delete from MST all edges with $c_e \leq 0$.

Remark. If G is not connected, add edges to it with cost $-M$, where $M > 0$ is large.

- The above algorithm will compute a max cost forest: Consider any two components of the computed forest. By the definition of MST algorithm, the edge deleted from the computed spanning tree has the smallest cost in the edges between the two components (w.r.t. $-c_e$), so all edges between this two components have negative costs. Also, for any edges e not connecting two components of the forest, if it has a positive cost, then it will be added to the computed spanning tree, hence a contradiction. Similar for the case when an edge is between two vertices of a component of the forest.
- We should have M greater than the largest absolute value of the negative c_e , so that when we are computing the MST, the "added" edges will never be selected.

1.3.4 Kruskal for Maximum Cost Forest

Algorithm 3

$H = (V, \emptyset)$.

while $\exists e : c_e > 0$, with endpoints in different connected components of H **do**
 $e =$ highest cost edge whose endpoints are in different components of H .

$H \leftarrow H + e$

end while

return H

To solve MST (alternatively):

- Add $-M$ to $c_e, \forall e$ such that $c_e - M < 0$
- Solve maximum cost forest w.r.t. $c'_e = -(c_e - M)$

If G is connect, and with all costs $c'_e > 0$, the above algorithm will find a spanning tree with the largest cost w.r.t. c'_e , that is, a spanning tree with the smallest cost w.r.t c_e .

2 Matroids

Look at edge sets of forests, i.e. instead of finding $H = (V, F)$, we just refer to F .

Algorithm 4 Generic Greedy

```

 $F \leftarrow \emptyset.$ 
while  $\exists e : F \cup \{e\} \in I$  and  $c_e > 0$  do
    choose such  $e$  with largest  $c_e$ ;
     $F \leftarrow F \cup \{e\}$ 
end while
return  $F$ 

```

where I here represents the set of all forests.

2.1 Matroid 1

Definition 2.1: Matroids

Let S be a ground set. Let $I \subseteq 2^S$ (the set of all subsets of S). $M = (S, I)$ is called a Matroid if it satisfies the following:

(M1) $\emptyset \in I$

(M2) If $F \in I$, $F' \subseteq F$, then $F' \in I$.

(M3) For all $A \subseteq S$, every inclusionwise maximal element of I that is contained in A (definition of the basis of A) has the same cardinality. That is, $B \in I$ is a subset of A and no other subsets of A in I is a strict superset of B , then B is a basis of A .

Example 2.2

- Let $G = (V, E)$. Set $S = E, I = \{ \text{all forest} \}$. We get a **Graphical/Forest Matroid**.
- Let $S = \{1, \dots, n\}$. Let $r \in \{0, \dots, n\}$, $I = \text{set of all subsets of } S \text{ with at most } r \text{ elements}$. We have

$$U_n^r = (S, I) \implies \text{Uniform matroid of rank } r$$

Q1: Is U_n^r a matroid?

(M1) $|\phi| = 0 \leq r$

(M2) If $A \in I$ and $B \subseteq A$, then $|B| \leq |A| \leq r \implies B \in I$.

(M3) If there are two basis B_1, B_2 of A and $|B_1| < |B_2| \leq \min\{r, |A|\}$, then let $e \in B_2 \setminus B_1$. Then $B_1 \cup \{e\} \subseteq A$ and $|B_1 \cup \{e\}| \leq |B_2| \leq \min\{r, |A|\}$. So B_1 is not a basis, contradiction.

- Let N be an $m \times n$ matrix of real numbers. Let $S = \{1, \dots, n\}$. $I = \{A \subseteq S : \text{columns indexed by } A \text{ are linearly independent}\}$. We call this a **Linear Matroid**.
e.g.:

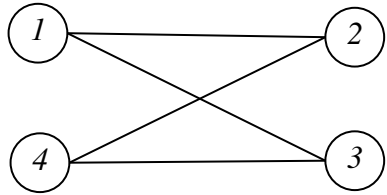
$$N = \begin{pmatrix} 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 0 & 2 & 0 \end{pmatrix}$$

, then $I = \{\emptyset, \{1\}, \dots, \{5\}, \{1, 2\}, \{1, 3\}, \{1, 5\}, \{2, 3\}, \{2, 4\}, \{2, 5\}, \dots\}$.

(M1) $\emptyset \in I$.

(M2) If a set of vectors is linearly independent, then any subset of it is also linearly independent.

(M3) Follows from linear algebra. (Note: basis of a vector space translate to basis of A).

Example 2.3

Let $S = V$, $I = \{A \subseteq V : A \text{ is a stable set}\}$.

(M1) $\emptyset \in I$

(M2) A subset of a stable set is also stable set.

(M3) Maximal (inclusionwise) subsets of S : $\{1, 4\}, \{2, 3\}$ which has the same cardinality. So is this example a matroid? **NO!**

$A = \{1, 2, 3\}$, then both $\{1\}, \{2, 3\}$ are the maximal subsets of A that are in I but they have different cardinality.

Definition 2.4: nomenclature of matroids

- Elements of I are called independent sets.
- Minimal dependent sets are called circuit (in the forest sense, cycle are circuits).
- If $M = (S, I)$ satisfies (M1), (M2), it's called an independence system.
- The rank of A : $r(A) := \max\{|B| : B \subseteq A, B \in I\}$
- The basis of M = the basis of S .
- $r(S)$ is the rank of M (matroid or independence system).
- $\rho(A) := \min\{|B| : B \text{ is a basis of } A\}$. Note:

$$M \text{ is a matroid} \iff \rho(A) = r(A), \forall A \subseteq S$$

2.2 Matroid 2

Maximum Weight independent Set (for independent systems):

Given $M = (S, I)$ independence system, $c_e \in \mathbb{R}_+$ (I can always delete the ones with $c_e < 0$, so assume $c_e \geq 0$), for all $e \in S$, find $A \in I$ maximizing $c(A) := \sum_{e \in A} c_e$.

```

 $F \leftarrow \emptyset$ 
while  $\exists e : F \cup \{e\} \in I$  and  $c_e > 0$  do:
    Choose such  $e$  with largest  $c_e$ ;
     $F \leftarrow F \cup \{e\}$ 
end while
return  $F$ 

```

Theorem 2.5: Rado, Edmonds

Let M be a matroid, $c \in \mathbb{R}_+^S$. Then greedy algorithm above finds Maximum Weight Independent Set.

Proof. Later □

Theorem 2.6: Rado, Edmonds

Let $M = (S, I)$ be an independence system. Then greedy finds an optimal independent set $\forall c \in \mathbb{R}_+^S$ if and only if M is a matroid.

Proof.

- ($\Leftarrow$) By Theorem 9 above.
- ($\Rightarrow$) Suppose M is not a matroid. Let $A \subseteq S$, A_1 and A_2 be bases of A with $|A_1| < |A_2|$. Let

$$c_e = \begin{cases} v_1, & \forall e \in A_1 \\ v_2, & \forall e \in A_2 \setminus A_1 \\ 0, & \forall e \notin A_1 \cup A_2 \end{cases}$$

Choose $v_1 > 0$ and $v_1 > v_2 > \frac{|A_1|}{|A_2|}v_1$. Then since all other elements have cost zero, the greedy algorithm only considers the elements in $A_1 \cup A_2$. Since $v_1 > v_2$, the algorithm will select A_1 first, since A_1 is a basis of A , the algorithm can't add more elements to it, so it stops and output A_1 . Then A_2 has cost $v_1|A_1 \cap A_2| + v_2|A_2 \setminus A_1| \geq v_2|A_2| > v_1|A_1|$. So the greedy algorithm does not output an optimal solution when all $c \in \mathbb{R}_+^S$, contradiction. □

Theorem 2.7: Jenkyns 176

Let (S, I) be an independent system. Let $gr_{S,I}$ be the total weight of the independent set formed by the greedy algorithm and $opt_{S,I}$ be the optimal solution weight. Then

$$gr_{S,I} \geq q(S, I) opt_{S,I}$$

where $q(S, I) = \min_{A \subseteq S, r(A) \neq 0} \frac{\rho(A)}{r(A)}$ (rank quotient).

Proof. Let $S = \{e_1, \dots, e_n\} : c_{e_1} \geq \dots \geq c_{e_n}$. Let $S_j := \{e_1, \dots, e_j\}$ and $S_0 := \emptyset$. Let $G \in I$ be solution obtained by greedy, $\sigma \in I$ be the optimal solution and $G_j = G \cap S_j$; $\sigma_j = \sigma \cap S_j$.

$$c(G) = \sum_{j \in G} c_j = \sum_{j=1}^n c_{e_j} (|G_j| - |G_{j-1}|) = \sum_{j=1}^n |G_j| \underbrace{(c_{e_j} - c_{e_{j+1}})}_{\Delta_j \geq 0}$$

note that if $e_j \in G$, then $|G_j| - |G_{j-1}| = 1$, otherwise, $|G_j| - |G_{j-1}| = 0$ and $c_{e_{n+1}} := 0$. Greedy computes a maximum independent subset of S_j implies G_j is a basis of S_j implies

$$\begin{aligned} c(G) &= \sum_{j=1}^n |G_j| \Delta_j \\ &\geq \sum_{j=1}^n \rho(S_j) \Delta_j \\ &\geq \sum_{j=1}^n q(S, I) r(S_j) \Delta_j \\ &\geq \sum_{j=1}^n q(S, I) |\sigma_j| \Delta_j \\ &= q(S, I) \sum_{j=1}^n |\sigma_j| (c_{e_j} - c_{e_{j+1}}) \\ &= q(S, I) \sum_{j=1}^n c_{e_j} (|\sigma_j| - |\sigma_{j-1}|) \\ &= q(S, I) \sum_{j \in \sigma} c_j \\ &= q(S, I) c(\sigma) \end{aligned}$$

□

Hence, by Jenkyn's results, we have if M is a matroid, greedy gets an optimal solution. And **Theorem 9 is proved by it.**

How fast is Greedy? Hence a total $O(|S|)$ times executed.

```

F ← ∅ O(1)
while  $\underbrace{\exists e : F \cup \{e\} \in I}_{\text{can be checked in time } Poly(|S|)?} \text{ and } c_e > 0 \text{ do :}$ 
```

Choose such e with largest c_e ; $O(|S|)$

$F \leftarrow F \cup \{e\}$ $O(1)$

end while

return F $O(1)$

2.3 Matroid 3

Theorem 2.8

Let $M = (S, I)$ independent system. Then $(M3) \iff (M3') : \forall X, Y \in I, |X| > |Y|, \exists x \in X \setminus Y : Y \cup \{x\} \in I$.

Proof.

- $(M3') \implies (M3)$ trivial.
- $(M3) \implies (M3')$. Let $X, Y \in I$ and $|X| > |Y|$. Consider $A = X \cup Y$. Then Y is not a basis of A because by $(M3)$, and $|X| > |Y|$, we have $|Y| < r(A)$. Then there exists $x \in A \setminus Y = X \setminus Y : Y \cup \{x\} \in I$.

□

Example 2.9

Let $G = (V, E), W \subseteq V$ a stable set. Let $k_v \in \mathbb{Z}_+, \forall v \in W, S = E, I = \{F \subseteq E : |\delta(v) \cap F| \leq k_v, \forall v \in W\}$. Clearly $(M1), (M2)$ hold.
 $(M3')$ Let $X, Y \subseteq E, X, Y \in I, |X| > |Y|$. Let $W_Y = \{v \in W : |\delta(v) \cap Y| = k_v\}$. Also, $2|X| = \sum_{v \in V} |X \cap \delta(v)|$. then

$$\begin{aligned} 2|X| &= \sum_{v \in W_Y} \underbrace{|X \cap \delta(v)|}_{\leq k_v} + \sum_{v \in W \setminus W_Y} |X \cap \delta(v)| + \sum_{v \in V \setminus W} |X \cap \delta(v)| \\ 2|Y| &= \sum_{v \in W_Y} \underbrace{|Y \cap \delta(v)|}_{= k_v} + \sum_{v \in W \setminus W_Y} |Y \cap \delta(v)| + \sum_{v \in V \setminus W} |Y \cap \delta(v)| \end{aligned}$$

Since $|X| > |Y|$, there exists $x \in X \setminus Y : x \in \delta(v)$ only for some $v \notin W_Y$. Otherwise, all $x \in X$ are either in Y or incident to W_Y , then $|X|$ is the number of edges in X incident to W_Y and the rest. While the rest part of X are all in Y but not incident to W_Y which are in the set of edges in Y but not incident to W_Y , and the number of edges in X incident to W_Y is less than or equal to number of edges in Y incident to W_Y . Mathematically, say K_X is subset of X such that $x \in K_X \iff x \in \delta(v)$ for some $v \in W_Y$. K_Y is the subset such that $y \in K_Y \iff y \in \delta(v)$ for some $v \in W_Y$. And $|K_X| \leq |K_Y|$ by the definition of W_Y . Then, $X \setminus K_X \subseteq Y \setminus K_Y$. Hence, $|X| \leq |Y|$, contradiction. Then $Y \cup \{x\}$ satisfies the condition.

2.3.1 Circuit characterization

Theorem 2.10: Circuit

If instead of describing I , you are given the set of circuits (min. dependent set) $(\mathcal{C})$ of M , then $A \in I \iff \nexists c \in \mathcal{C} : c \subseteq A$.

Proof.

- $\implies$: by (M2), any subset of A should be in I , so it has no subset in $\mathcal{C}$.
- $\impliedby$: Suppose A is not in I , then its dependent, keep deleting elements from A till it's in $\mathcal{C}$, then we have a subset of A which is in $\mathcal{C}$, contradiction.

□

Example 2.11

$S = \{1, 2, 3, 4\}$, $\mathcal{C} = \{\{4\}, \{1, 2, 3\}\}$, $I = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}\}$.

Q: When is $\mathcal{C} \subseteq 2^S$ the set of circuits of a matroid?

Theorem 2.12

Let $M = (S, I)$ be a matroid. Then $\forall A \in I$, $\forall e \in S$, $A \cup \{e\}$ contains at most 1 circuit.

Proof. Let A be smallest set so that

- $A \in I$
- $\exists e : A \cup \{e\}$ has two distinct circuits C_1, C_2 .

Note $e \in C_1 \cap C_2$, otherwise, A has a circuit then it can't be in I .

By the choice of A , we have $A \cup \{e\} = C_1 \cup C_2$ (otherwise there exists $u \in A \setminus (C_1 \cup C_2)$, then $A \setminus \{u\}$ is a smaller set satisfying the properties above).

Since $C_1 \not\subseteq C_2, C_2 \not\subseteq C_1$ (if $C_1 \subset C_2$, then C_2 is not a circuit), let $e_1 \in C_1 \setminus C_2$ and $e_2 \in C_2 \setminus C_1$. Consider $A' = (C_1 \cup C_2) \setminus \{e_1, e_2\}$, if A' has a circuit C , then

- $C \neq C_1, C \neq C_2$ because $C \cap \{e_1, e_2\} = \emptyset$.
- Since $e_1 \notin C_2, C_2 \subseteq \{A \setminus e_1\} \cup \{e\}$, similarly, C is also a subset of it.
- Then $A \setminus \{e_1, e_2\}$ will be a set satisfying the properties, contradicts to the minimality of A , so A' has no circuit, so $A' \in I$.

so A, A' are bases of $C_1 \cup C_2$, with $|A'| < |A|$ which contradicts to M being a matroid. □

Theorem 2.13

Let $\mathcal{C} \subseteq 2^S$. Then $\mathcal{C}$ is the set of circuits of a matroid iff

(C1) $\emptyset \notin \mathcal{C}$

(C2) If $C_1, C_2 \in \mathcal{C}, C_1 \subseteq C_2$, then $C_1 = C_2$.

(C3) If $C_1, C_2 \in \mathcal{C}, C_1 \neq C_2$ and $e \in C_1 \cap C_2$, then there exists $C \in \mathcal{C}$ with $C \subseteq (C_1 \cup C_2) \setminus \{e\}$.

Proof.

- ($\implies$): $(C_1), (C_2)$ are trivial to prove. Suppose (C_3) not true, then $A := (C_1 \cup C_2) \setminus \{e\} \in I$ by $\nexists c \in \mathcal{C}$ such that $C \subseteq A$. This implies $A \cup \{e\}$ has two distinct circuits, contradicts to the previous theorem.
- Define $I = \{A \subseteq S : \nexists C \in \mathcal{C} \text{ with } C \subseteq A\}$. Let $M = (S, I)$, then $(M1), (M2)$ clearly hold.
 Suppose $(M3)$ is false, let A_1, A_2 be the bases of $A \subseteq S$ with $|A_1| < |A_2|$, choose A_1, A_2 with largest $|A_1 \cap A_2|$. Let $e \in A_1 \setminus A_2$ (it exists because $A_1 \not\subseteq A_2$) and $A_2 \cup \{e\}$ contains a circuit C . If $A_2 \cup \{e\}$ contains $C' \neq C$ (note $e \in C \cap C'$), then $(C_3) \implies A_2$ contains a circuit, but $A_2 \in I$. Hence C is a unique circuit in $A_2 \cup \{e\}$. Let $f \in C \setminus A_1 \implies \underbrace{(A_2 \cup \{e\}) \setminus \{f\}}_{A_3} \in I$,
 but $|A_3 \cap A_1| > |A_2 \cap A_1|, |A_3| = |A_2| > |A_1|$, contradiction. Note: we can make A_3 a basis by adding elements, but the inequalities above still hold.

□

2.4 Matroid 4

Theorem 2.14: Bases characterization

Instead of giving I , we are given $\mathbb{B}$, the set of bases of M , then $A \in I \iff A \subseteq B$, for some $B \in \mathbb{B}$.

Theorem 2.15

Let $\mathbb{B} \subseteq 2^S$. $\mathbb{B}$ is the set of bases of a matroid (S, I) if and only if

(B1) $\mathbb{B} \neq \emptyset$

(B2) $\forall B_1, B_2 \in \mathbb{B}, x \in B_1 \setminus B_2$, there exists $y \in B_2 \setminus B_1$ such that $(B_1 \setminus \{x\}) \cup \{y\} \in \mathbb{B}$.

Theorem 2.16

Let $\mathbb{B} \subseteq 2^S$. $\mathbb{B}$ is the set of bases of a matroid (S, I) if and only if

(B1) $\mathbb{B} \neq \emptyset$

(B2) $\forall B_1, B_2 \in \mathbb{B}, y \in B_2 \setminus B_1$, there exists $x \in B_1 \setminus B_2$ such that $(B_1 \setminus \{x\}) \cup \{y\} \in \mathbb{B}$.

2.5 Polymatroids

Let $M = (S, I)$ be a matroid, $c \in \mathbb{R}_+^S$. Let $x \in \mathbb{R}^S$ be decision variables.

$$\begin{aligned}
 (P_M) \quad & \max c^T x \\
 & \text{s.t. } x(A) \leq r(A), \forall A \subseteq S \\
 & x \geq 0
 \end{aligned}$$

Note: If $J \in I$, then x^J (incidence vector) is feasible for (P_M) .

Theorem 2.17

Let $M = (S, I)$ be a matroid, and let G be the solution returned by the greedy algorithm. Then x^G is optimal for (P_M) .

Definition 2.18

A function $f : 2^S \mapsto \mathbb{R}$ is called submodular if $\forall A, B \subseteq S$,

$$f(A) + f(B) \geq f(A \cap B) + f(A \cup B)$$

Proposition 2.19

Let $M = (S, I)$. Then $r(A)$ is submodular.

Proof. Let $A, B \subseteq S$. Let $J_\cap$ be a basis of $A \cap B$ (i.e. $|J_\cap| = r(A \cap B)$). Extend $J_\cap$ to a basis J_B of B (i.e. $|J_B| = r(B)$) (by keep adding elements of B to $J_\cap$ until we get a maximal independent set contained in B). Similarly, extend J_B to a basis $J_\cup$ of $A \cup B$ (i.e. $|J_\cup| = r(A \cup B)$). Let $J' = J_\cup \setminus (J_B \setminus J_\cap)$

- Since $J' \subseteq J_\cup$, we have $J' \in I$.
- Suppose there exists $v \in J' \setminus A$, then $v \in J_\cup \setminus A$ and $v \notin J_B \setminus J_\cap$. Since $v \notin A$, we have $v \notin J_\cap$. so $v \notin J_B$, and $v \in B$. Since $J_\cup$ is a basis, $J_B \cup \{v\} \in I$, then J_B is not a basis of B , contradiction. So $J' \subseteq A$.

Thus:

$$r(A) + r(B) \geq |J'| + |J_B| = |J_\cup| - (|J_B| - |J_\cap|) + |J_B| = |J_\cup| + |J_\cap| = r(A \cup B) + r(A \cap B)$$

□

Definition 2.20

Let $f : 2^S \mapsto \mathbb{R}_+$ be submodular, then

$$\{x \in \mathbb{R}^S : x(A) \leq f(A), \forall A \subseteq S\}$$

is called a Polymatroid.

Note: May assume $f(\emptyset) = 0$, f is monotone (i.e. $X \subseteq Y \subseteq S \iff f(X) \leq f(Y)$). Consider (where f monotone and $f(\emptyset) = 0$)

$$(P_f) \quad \begin{aligned} & \max c^T x \\ & \text{s.t. } x(A) \leq f(A), \forall A \subseteq S \\ & \quad x \geq 0 \end{aligned}$$

$$\begin{aligned}
& \min \sum_{A \subseteq S} f(A) y_A \\
(D_f) \quad & s.t. \sum_{A: e \in A} y_A \geq c_e, \forall e \in S \\
& y \geq 0
\end{aligned}$$

Primal Greedy

$S = \{e_1, \dots, e_n\}$, $c_{e_1} \geq \dots \geq c_{e_k} \geq 0 \geq c_{e_{k+1}} \geq \dots \geq c_{e_n}$. $S_j = \{e_1, \dots, e_j\}$ and

$$x_{e_j} = \begin{cases} f(S_j) - f(S_{j-1}), & \forall j = 1, \dots, k \\ 0, & \forall j > k \end{cases}$$

If $f(S_j) = r(S_j)$, for $M = (S, I)$ matroid. Let G be a greedy solution, $G_j := G \cap S_j$. If G_{j-1} is a basis of S_{j-1} , then when $r(S_j) = r(S_{j-1})$, we have $x_{e_j} = 0$, so $e_j \notin G_j$. Then $G_j = G_{j-1}$, so G_{j-1} is also a basis of S_j . When $r(S_j) > r(S_{j-1})$, $x_{e_j} = 1$, so $e_j \in G \implies G_j = G_{j-1} \cup \{e_j\} \implies G_j$ is a basis of S_j .

Dual Greedy

$$\begin{aligned}
y_{S_j} &= c_{e_j} - c_{e_{j+1}}, \forall j = 1, \dots, k-1 \\
y_{s_k} &= c_{e_k} \\
y_A &= 0, \text{ for all other } A
\end{aligned}$$

We can show that x, y above are optimal solutions by Complementary Slackness conditions.

Corollary 2.21

Let $M = (S, I)$, $c \in \mathbb{R}^S$, $J \in I$. Then J is an inclusionwise minimal, max weight independent set if and only if

- (a) $e \in J \implies c_e > 0$
- (b) $e \notin J, J \cup \{e\} \in I \implies c_e \leq 0$
- (c) $e \notin J, f \in J, (J \cup \{e\}) \setminus \{f\} \in I \implies c_e \leq c_f$.

Proof.

- $\implies$: trivial
- $\impliedby$: Consider (P_r) , where r is the rank function of M which is monotone, submodular and $r(\emptyset) = 0$. (note J is independent, so J is feasible for P_r). Let y be the solution from greedy. Let x^J be the characteristic vector of J . Then

$$\sum_{A: e_j \in A} y_A = c_{e_j}, \forall j \leq k$$

and

$$a) \implies x^J e_j = 0, \forall j > k$$

Thus for all $j \in \{1, \dots, n\}$, we have $x_{e_j}^J = 0$ OR $\sum_{A: e_j \in A} y_A = c_{e_j}$.

Pick $y_A > 0$. By construction, $A = S_j$ for $j \leq k$. Note $x^J(S_j) = |J \cap S_j| = |J_j|$. Suppose $|J_j| < r(S_j)$, then J_j is not a basis of S_j , but it's an independent set so there exists $e \in S_j \setminus J$ such that $J_j \cup \{e\} \in I$.

Case 1 $J \cup \{e\} \in I$, then $b \implies c_e \leq 0$, but $e \in S_j \implies c_e > 0$, contradiction.

Case 2 $J \cup \{e\} \notin I$. Extend $J_j \cup \{e\}$ to a basis J' of $J \cup \{e\}$. Note J is a basis of $J \cup \{e\}$. Hence, $|J'| = |J|$ by both being basis of $J \cup \{e\}$.

Then there exists $f \in J \setminus S_j$ such that $J' = (J \cup \{e\}) \setminus \{f\} \in I$. This f exists because $e \in J' \setminus J$, so there is $f \in J \setminus J'$, and $J_j \subseteq J \cap J'$, so $f \notin J_j$, which implies $f \notin S_j$.

Then by c), $c_e \leq c_f$.

By $y_{S_j} = c_{e_j} - c_{e_{j+1}} > 0 \implies c_{e_j} > c_{e_{j+1}}$ and $f \notin S_j \implies c_{e_{j+1}} \geq c_f \geq c_e$, we have $c_{e_j} > c_{e_{j+1}} \geq c_f \geq c_e$, but $e \in S_j$, so $c_e \leq c_{e_j}$, contradiction.

Hence, $x^J(S_j) = r(S_j)$.

Hence, Complementary Slackness conditions hold, so x^J is optimal for (P_r) which implies J is a maximal weight independent set. And a) implies the inclusionwise minimality. $\square$

2.6 Matroid Construction

Let $M = (S, \mathcal{I})$ be a matroid.

1. Deletion: Let $B \subseteq S$, $M \setminus B := (S', \mathcal{I}')$ is a matroid, where $S' := S \setminus B$, $\mathcal{I}' := \{A \subseteq S \setminus B : A \in \mathcal{I}\}$.
2. Truncation: Let $k \in \mathbb{Z}_+$, define $\mathcal{I}' := \{A \in \mathcal{I} : |A| \leq k\}$. Then $M' = (S, \mathcal{I}')$ is a matroid.
3. Disjoint Union: Let $M_i = (S_i, \mathcal{I}_i)$, $\forall i \in \{1, \dots, k\}$ be matroids, with $S_i \cap S_j = \emptyset$, $\forall i \neq j$. Then $M_1 \oplus \dots \oplus M_k = (S, \mathcal{I})$, with $S = \cup_{i=1}^k S_i$. And $\mathcal{I} = \{A \subseteq S : A \cap S_i \in \mathcal{I}_i, \forall i = 1, \dots, k\}$ is a matroid.

Proof.

(M1) hold

(M2) hold

(M3) Let B be a basis of $A \subseteq S$. Let $B_i = B \cap S_i$, $\forall i$. Then $B_i \in \mathcal{I}$, but also, it is a basis of $A \cap S_i$. (otherwise, we can add $\alpha \in A \cap S_i$ to B_i , hence to B , then B is not a basis of A). This implies $|B| = \sum_{i=1}^k |B_i| = \sum_{i=1}^k r_i(A \cap S_i)$, thus all basis of A have the same size.

$\square$

Example: Partition Matroid: Let $S = S_1 \dot{\cup} S_2 \dots \dot{\cup} S_k$, $b_1, \dots, b_k \in \mathbb{Z}_+$. $M = (S, \mathcal{I})$ where $\mathcal{I} = \{A \subseteq S : |A \cap S_i| \leq b_i, \forall i = 1, \dots, k\}$. Then $M_i = (S_i, \mathcal{I}_i)$, $\mathcal{I}_i = \{J \subseteq S_i : |J| \leq b_i\}$ is the uniform matroid. And $M = M_1 \oplus \dots \oplus M_k$.

4. **Contraction:** Let $B \subseteq S$, let J be a basis of B . Then $M/B = (S', \mathcal{I}')$ where $S' = S \setminus B$, $\mathcal{I}' = \{A \subseteq S' : A \cup J \in \mathcal{I}\}$.

Proposition 2.22

If M is forest matroid of $G = (V, E)$, $B \subseteq E$, then M/B is a forest matroid of G/B (contraction in graph theory).

Theorem 2.23

M/B is a matroid independent from choice of J , and its rank fcn is $r_{M/B}(A) = r_M(A \cup B) - r_M(B)$.

Proof. (M1), (M2) clearly hold. (M3): Let $A \subseteq S' = S \setminus B$, let J' be an M/B basis of $A \implies J \cup J' \in \mathcal{I}$.

Claim. $J \cup J'$ is an M -basis of $A \cup B$.

Proof. Suppose not, then there is $e \in A \cup B \setminus J \cup J'$ and $J \cup J' \cup \{e\} \in \mathcal{I}$. If $e \in B$, then $J \cup \{e\} \in \mathcal{I}$ and it's a subset of B , contradicts to J being a basis of B . If $e \notin B$ then $e \in A \setminus B$, then $(J' \cup \{e\}) \cup J \in \mathcal{I} \implies J' \cup \{e\} \in \mathcal{I}'$, contradicts to J' being a basis of A . $\square$

By the claim, $|J \cup J'| = r_M(A \cup B) \implies |J| + |J'| = r_M(A \cup B) \implies |J'| = r_M(A \cup B) - |J| = r_M(A \cup B) - r_M(B)$. Thus, $A \in \mathcal{I}'$ if and only if $|A| = r_{M/B}(A) = r_M(A \cup B) - r_M(B)$ which doesn't depend on J . $\square$

5. **Duality:** $M^* = (S, \mathcal{I}^*)$, $\mathcal{I}^* = \{A \subseteq S : S \setminus A \text{ has a basis of } M\} = \{A \subseteq S : r_M(S \setminus A) = r_M(S)\}$. Note here the basis of M means the basis of S .

Example: $M = U_n^r$, $A \subseteq S = \{1, \dots, n\}$, $A \in \mathcal{I}^* \iff |A| \leq n - r \implies M^* = U_n^{n-r}$.

Theorem 2.24

M^ is a matroid with rank function $r^*(A) = |A| + r_M(S \setminus A) - r_M(S)$.*

Proof. Clearly (M1), (M2) hold. For (M3), let $A \subseteq S$, let J^* an M^* -basis of A . Let J be an M -basis of $S \setminus A$. Extend J to an M -basis J' of $S \setminus J^*$. By definition, we know J' is an M -basis of S .

Claim. $A \setminus J^* \subseteq J'$

Proof. Suppose $e \in (A \setminus J^*) \setminus J' \implies J' \subseteq S \setminus (J^* \cup \{e\})$, and since J' is an M -basis of S , $J^* \cup \{e\} \in I^*$, contradiction. $\square$

Then

$$|J'| = |A \setminus J^*| + |J| = |A| - |J^*| + |J| \iff |J^*| = |A| - |J'| + |J| = |A| - r_M(S) + r_M(S \setminus A)$$

$\square$

3 Matchings

3.1 Matchings 1

Definition 3.1

Given a graph $G = (V, E)$, a subset $M \subseteq E$ is a matching if $|\delta(v) \cap M| \leq 1, \forall v \in V$; i.e., every vertex incident to at most one edge in M .

Given a matching M , a vertex v is called M -covered if $|\delta(v) \cap M| = 1$, and it's called M -exposed otherwise.

Note: There are $2|M|$ M -covered vertices and $|V| - 2|M|$ M -exposed vertices.

- A matching is perfect if there are no M -exposed vertices.
- The size of the largest cardinality matching in G will be denoted as $\nu(G)$. (M is a perfect matching if and only $\nu(G) = \frac{|V(G)|}{2}$.)
- Given $G = (V, E)$, and a matching M , a path $P = (v_1, \dots, v_k)$ is called M -alternating if $\{v_{i-1}, v_i\} \in M \iff \{v_i, v_{i+1}\} \notin M, \forall i = 2, \dots, k-1$.
- An M -alternating path is called M -augmenting if v_1, v_k are exposed.
- Given $F_1, F_2 \subseteq E$, the symmetric difference between F_1, F_2 is defined as

$$F_1 \triangle F_2 = \{e \in E : e \text{ is in exactly one of } F_1, F_2\}$$

Theorem 3.2

Let M be a matching of $G = (V, E)$. Then M is a max cardinality matching if and only if there does not exist an M -augmenting path.

Proof.

- ($\implies$) Suppose there exists an M -augmenting path $P = \{v_0, \dots, v_k\}$. Let $e_i = \{v_{i-1}, v_i\}$, $\forall i = 1, \dots, k$. Let $M' = M \triangle E(P)$. Note since P is M -augmenting, we know v_0, v_k are M -exposed, so $e_1, e_k \notin M$, so k is odd. That is $|M'| = |(E(P) \setminus M) \cup (M \setminus E(P))| = |M| - |(E(P) \cap M)| + |(E(P) \cap M)| + 1 = |M| + 1$. Suppose M' is not a matching. Then there are two edges in M' incident to one vertex. If both edges are in M , then M is not a matching, contradiction. Hence, at least one of them is in $M' \setminus M$, call it e . Then e is in $E(P) \setminus M$. If we have $v_i v_{i+1} e v_{i+2} v_{i+3}$. Then $v_i v_{i+1}$ and $v_{i+2} v_{i+3}$ are not in M' by they are in both M and $E(P)$, so v_{i+1}, v_{i+2} are not M' -exposed, contradiction. If one end of e is v_0 (v_k), then if v_1 incident to another edge in e' , we know $e' \neq v_1 v_2$ by $v_1 v_2 \in M \implies v_1 v_2 \notin M'$, so $e' \notin E(P)$, so $e' \in M$, then v_1 incidence to e' and $v_1 v_2$ in M , contradiction. Hence, v_0 is incidence to another e' in M' , then $e' \notin E(P) \implies e' \in M$, but v_0 is M -exposed, contradiction. Hence, M' is a matching, and $|M'| > |M|$, contradiction, so not such P exists.

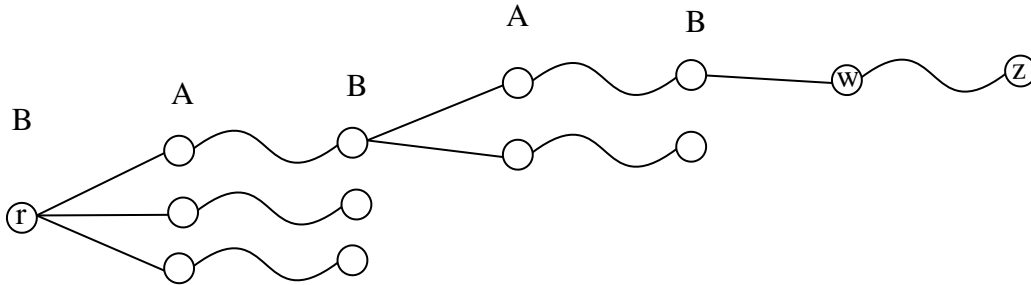
- ($\Leftarrow$) Suppose M' is a matching of G with $|M'| > |M|$. Consider $G' = (V, M \Delta M')$. Note $|\delta_{G'}(v)| \leq 2, \forall v \in V$, because if $|\delta_{G'}(v)| = 3$ for some $v \in V$, then there are three edges incident to it, so there are at least two edges in the same matching incident to it, contradiction. Also $|\delta_{G'}(v)| \leq 2 \implies G'$ is a (edge) disjoint union of paths and cycles, and all of them are alternating (w.r.t. M and M'). Also note if C is a cycle in G' , $|E(C) \cap M| = |E(C) \cap M'|$, otherwise C is an odd cycle and there will be a vertex incident to two edges in M or M' , contradiction. Hence, there exists a path P with $|E(P) \cap M'| > |E(P) \cap M|$, then P is the desired M -augmenting path in G , contradiction.

□

- Q: Does there exist a path from a vertex u to a vertex v ?
A: Use Breadth First Search.
- Q: Does there exist an M -alternating path from an M -exposed vertex u to an M -exposed vertex v ?
A: Similar, keep the path you are looking for alternating. Instead of constructing a Breadth First Search Tree, we construct an "alternating" trees. It can keep track of nodes at odd/even distance from the tree root.

Tentative Algorithm:

Input: $G = (V, E)$, M is a matching, $r \in V$ is M -exposed. $T \leftarrow (\{r\}, \emptyset)$, $A(T) \leftarrow \emptyset$, $B(T) \leftarrow \{r\}$, where A represents the node at odd distance from the tree root, and B represents the node at even distance from the tree root.



In the tree, we use tilde lines to represent the edges in M and straight lines otherwise. Also, we can see that each path from the root r to a node in T is an M -alternating path in G .

Case 1: If we can find $vw \in E$: $v \in B(T)$, $w \notin V(T)$, and w is M -covered. We can extend T using vw .

Let $z \in V$: $wz \in M$, since v is in T , v is incident to another edge in M , so $z \neq v$. Then update $V(T) \leftarrow V(T) \cup \{w, z\}$, $B(T) \leftarrow B(T) \cup \{z\}$, $A(T) \leftarrow A(T) \cup \{w\}$, $E(T) \leftarrow E(T) \cup \{vw, wz\}$.

Case 2: If we find $vw \in E$: $v \in B(T)$, $w \notin V(T)$ and w is M -exposed, then we find an M -augmenting path from r to w , which is $P' = P + vw$, where P is the M -alternating path from r to v in T , $M \leftarrow M \Delta P'$.

Hence, the tentative algorithm can be written as

Algorithm 5 Tentative Algorithm for Matchings

$G = (V, E)$, M is a matching, $r \in V$ and it's M -exposed. $T \leftarrow (\{r\}, \emptyset)$, $A(T) \leftarrow \emptyset$, $B(T) \leftarrow \{r\}$ (initialized T with r).

while $\exists vw \in E : v \in B(T), w \notin V(T)$ **do**:

if w is M -covered **then**

 Use vw to extend T

else

 Use vw to augment M ;

if $\exists$ M -exposed vertex $r \in V$ **then**

 Initialize T with r

else

 Stop

end if

end if

end while

return M

This does not always work (e.g. G is not connected).

3.2 Matching 2

Definition 3.3

A graph is bipartite if there exists a partition (A, B) of V such that $\forall e \in E, |e \cap A| = |e \cap B| = 1$.

Theorem 3.4: Hall's Theorem

Let $G = (V, E)$ be bipartite, with bipartition $V = A \dot{\cup} B$. Then there exists a matching covering A if and only if $|N(X)| \geq |X|, \forall X \subseteq A$, where $N(X) := \{v \in V \setminus X : \exists u \in X \text{ with } \{u, v\} \in E\}$.

Proof.

- ($\implies$) If there exists $X \subseteq A : |X| > |N(X)|$, then since vertices only matched to vertices in $N(X)$, no matching can cover all vertices in X (there is a vertex in X having no neighbors).
- ($\impliedby$) By induction, cases $|A| = 0, |A| = 1$ are trivial.
If $|N(X)| - |X| > 0, \forall X \subset A, X \neq \emptyset$, pick $uv \in E$ with $u \in A, v \in B$, and consider $G' = G \setminus \{u, v\}$, bipartite with bipartition $A' = A \setminus \{u\}, B' = B \setminus \{v\}$. Now, $\forall X \subseteq A', |N_{G'}(X)| \geq |N_G(X)| - 1 \implies |N_{G'}(X)| - |X| \geq 0, \forall X \subseteq A'$. By induction, there exist a matching M' covering A' , then $M' \cup \{u, v\}$ covers A .

If $|N(X)| = |X|$, for some $X \subset A, X \neq \emptyset$. By induction, there exist a matching M_1 in $G[X \cup N(X)]$ covering X . Now consider $G' = G[(A \setminus X) \cup (B \setminus N(X))]$. Note $\forall Y \subseteq A \setminus X$,

$$|N_{G'}(Y)| = |N_G(Y) \setminus N_G(X)| = |N_G(X \cup Y)| - \underbrace{|X|}_{=|N_G(X)|} \geq |X \cup Y| - |X| = |Y|$$

Hence, there exists a matching in G' covering $A \setminus X$, combine it with M_1 , there is a matching covering A . □

Corollary 3.5

Let $G = (V, E)$ be bipartite with bipartition $V = A \dot{\cup} B$. Then G has a perfect matching if and only if $|A| = |B|$ and $|X| \leq |N(X)|, \forall X \subseteq A$.

Algorithm 6 Algorithm for Perfect Matchings of bipartite graphs

$G = (V, E)$ be bipartite, initialize T with r .

while $\exists vw \in E : v \in B(T), w \notin V(T)$ **do**:

if w is M -covered **then**

 Use vw to extend T

else

 Use vw to augment M ;

if $\exists$ M -exposed vertex $r \in V$ **then**

 Initialize T with r

else

 Stop, output perfect matching M

end if

end if

end while

Output No Perfect Matchings exists. (*)

If algorithm reaches (*), then G has no perfect matching.

Proof. If the algorithm reaches (*), then

- $N(B(T)) = A(T)$. First, $A(T) \subseteq N(B(T))$, and if there exist a vertex $u \in N(B(T)) \setminus A(T)$ which is a neighbor of $v \in B(T)$, then $u \notin B(T)$, because otherwise, both u, v are at even distance from the root, and by G being bipartite, that means both u, v are in the same partition of G , and they are incident, contradiction.
- $|B(T)| > |A(T)|$. Suppose the tree has a leaf in $A(T)$, then by our algorithm, if it's M -exposed, we augment M , otherwise, we extend T , so all leaves of T are in $B(T)$. That is, for every vertex in $A(T)$, it has a neighbor in $B(T)$ in the tree with one larger height from the root, and since $r \in B(T)$, we have $|B(T)| > |A(T)|$.

- By what's above, we know $|B(T)| > |N(B(T))|$, by the Corollary above, G has no perfect matching.

□

Definition 3.6

$U \subseteq V$ is a vertex cover if $\forall e \in E, |e \cap U| \geq 1$. We let $\tau(G)$ be the size of the smallest cardinality vertex cover. Fact: $\nu(G) \leq \tau(G)$. Otherwise, consider the max cardinality matching, you need at least $|M|$ vertices to cover the M -covered vertices because for each edge in M , you need one of the ends in the vertex cover.

Theorem 3.7: König's Theorem

Let G be bipartite, then $\nu(G) = \tau(G)$.

3.3 Matching 3

Recall $\nu(G) \leq \tau(G)$ and equality holds for bipartite graph. Suppose $A \subseteq V$, let $H_1, \dots, H_k$ be odd connected components of $G \setminus A$.

Q: How many M -exposed vertices can there be?

If H_i has no M -exposed vertices, then there exists at least one edge in M from H_i to A (because there are odd number of vertices in H_i). But there are at most $|A|$ such edges, implies there are at least $k - |A|$ M -exposed vertices for all matching M .

Recall: there are $|V| - 2|M|$ M -exposed vertices in any matching, which implies $|V| - 2|M| \geq k - |A|, \forall M$. It is equivalent to $|M| \leq \frac{1}{2}(|V| - k + |A|)$. Then, let $k = oc(G \setminus A)$ (number of odd components of $G \setminus A$),

$$\nu(G) \leq \frac{1}{2}(|V| - oc(G \setminus A) + |A|), \forall A \subseteq V$$

We also note that if A is a vertex cover, then $G \setminus A$ is a graph with no edges, so $oc(G \setminus A) = |V| - |A|$, then the bound above becomes $|A|$.

Theorem 3.8: Tutte-Berge Formula

Let $G = (V, E)$ be a graph. Then

$$\max\{|M| : M \text{ is a matching}\} = \frac{1}{2} \min\{|V| - oc(G \setminus A) + |A| : A \subseteq V\}$$

Theorem 3.9: Tutte's Matching Theorem

G has a perfect matching $\iff oc(G \setminus A) \leq |A|, \forall A \subseteq V$.

Proof. If $oc(G) > 0$, then G has no perfect matching and $A = \emptyset$ violates $oc(G \setminus A) \leq |A|$.

If $oc(G) = 0$, then

G has a perfect matching

$$\begin{aligned} &\iff \nu(G) = \frac{n}{2} \\ &\iff n = \min\{n - oc(G \setminus A) + |A| : A \subseteq V\} \\ &\iff \min\{|A| - oc(G \setminus A) : A \subseteq V\} = 0 \end{aligned}$$

But for $A = \emptyset$, $|A| - oc(G \setminus A) = 0$, so 0 can be obtained, that is,

$$\min\{|A| - oc(G \setminus A) : A \subseteq V\} = 0 \iff oc(G \setminus A) \leq |A|, \forall A \subseteq V$$

□

So Tutte's Matching Theorem is proved by using Tutte-Berge Formula, which is what we want to prove now. Before that, we say $u \in V$ is essential if u is M -covered in EVERY maximum cardinality matching M ; otherwise, it is inessential.

Proof. of Tutte-Berge Formula.

Goal: Show a matching M and $A \subseteq V$ with exactly $oc(G \setminus A) - |A|$ vertices (which is saying $oc(G \setminus A) - |A| = |V| - 2|M|$). If such M, A are found, then the Tutte-Berge formula is proved. As we have shown before, $\nu(G) \leq \frac{1}{2}(|V| - oc(G \setminus A) + |A|)$, $\forall A \subseteq V$, so $\nu(G) \leq \frac{1}{2} \min\{|V| - oc(G \setminus A) + |A| : A \subseteq V\}$. When $oc(G \setminus A) - |A| = |V| - 2|M|$ many M -exposed vertices, we know

$$\nu(G) \geq |M| = \frac{1}{2}(|V| - oc(G \setminus A) + |A|) \leq \frac{1}{2} \min\{|V| - oc(G \setminus A) + |A| : A \subseteq V\}$$

so the Tutte-Berge Formula holds.

Now we do induction on $m = |E|$

Base: $m = 0$, let $A = \emptyset$, we are done. Now assume $m \geq 1$ and pick $uv \in E$:

Case 1: v is essential. Let $G' = G \setminus v$, then $\nu(G') < \nu(G)$. By induction, there exists matching M' in G' and $A' \subseteq V \setminus \{v\}$ with

$$|M'| = \frac{1}{2}(n - 1 - oc(G' \setminus A') + |A'|)$$

Let M be a matching of G with $|M| = \nu(G)$. Pick $e \in \delta(v) \cap M$ (it exists by v being essential). Then $\overline{M} = M \setminus e$ is a matching in G' which implies $|\overline{M}| = |M| - 1 \leq |M'|$. Now, suppose $|M| - 1 < |M'|$, then $|M| \leq |M'|$, then since M' is also a matching in G , we have $|M| \geq |M'|$, so $|M| = |M'|$, then M' is a maximum cardinality matching in G without v , so v is not essential, contradiction. Hence, $|M| - 1 = |M'|$. Then let $A = A' \cup \{v\}$, then $|A| = |A'| + 1$ and $G \setminus A = G' \setminus A'$, so

$$|M| = |M'| + 1 = \frac{1}{2}(n + 1 - oc(G' \setminus A') + |A'|) = \frac{1}{2}(n - oc(G \setminus A) + |A|)$$

we are done.

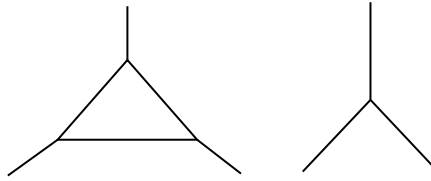
Case 2: u, v both are inessential. Later.

Let C be an odd cycle, let $G' = G/C$ (contracting C). That is $V'(G) = V(G) \setminus V(C) \cup \{\mathcal{C}\}$; $E(G') = \{e \in E(G) : e \cap C = \emptyset\} \cup \{v\mathcal{C} : \exists uv \in E(G), u \in V(C), v \notin V(C)\}$. Note from this point, we allow parallel edges. The idea is that a matching in G' can be extend to a matching in G with the same number of exposed vertices. The process is, let all edges in the matching of G' be in the matching of G , then let one vertex in C to represent the $\mathcal{C}$ in G' , and C has even number of vertices left, then choose edges so they are all M -covered.

Proposition 3.10

Let $G = (V, E)$, C an odd cycle, $G' = G/C$. Let M' a matching in G' . Then there exists a matching M of G such that the number of M -exposed vertices in G equals the number of M' -exposed vertices in G' .

Note we add $\frac{|C|-1}{2}$ new edges to M' to get M . Therefore, $\nu(G) \geq \nu(G') + \frac{|C|-1}{2}$, but the equality does not necessarily hold, for example



where the left graph G has $\nu(G) = 3$, the right one has $\nu(G') = 1$ and $\frac{|C|-1}{2} = 1$. An odd cycle is tight if $\nu(G) = \nu(G') + \frac{|C|-1}{2}$.

Now back to the proof, we pick a tight cycle C containing uv and where $\mathcal{C}$ is inessential in $G' = G/C$. Then there exist M' matching of G' , $A' \subseteq V(G')$:

$$|M'| = \frac{1}{2}(|V(G') - oc(G' \setminus A')| + |A'|)$$

If $\mathcal{C} \notin A'$, then any component of $G' \setminus A'$ containing $\mathcal{C}$ will be a component of $G \setminus A$ of same pairing after extending back (that is, if the component in G' is odd, then the component in G will also be odd because there are even number of vertices if deleting $\mathcal{C}$, and C has odd number of vertices, same if the component in G' is even). Hence, there are

$$oc(G' \setminus A') - |A'| = oc(G \setminus A) - |A| = |V| - |C| + 1 - 2|M'| = |V| - |C| + 1 - 2\left(|M| - \frac{|C|-1}{2}\right) = |V| - 2|M|$$

many M -exposed vertices.

Q: But why does such C exist? What if $\mathcal{C} \in A'$? □

3.4 Matching 4

Lemma 3.11

Let $uv \in E$. If u, v are inessential, then there is a tight odd cycle C containing the edge uv , such that $\mathcal{C}$ is inessential in $G' = G/C$.

Proof. Let M_u, M_v be maximum cardinality matchings exposing u, v respectively. (Note1: $uv \notin M_u \cup M_v$; Note 2: M_u, M_v covers v, u respectively by the maximality). Then

- Degree of u, v is 1 in $M_u \triangle M_v := F$ ((V, F) is a vertex disjoint union of M_u, M_v alternating paths/cycles).
- There exists an alternating path P starting at u and the other end z is M_v -exposed. Suppose the other end is M_u -exposed, then the path P is an M -augmenting path, contradicts to the maximality of M in G . If $z \neq v$, then $vu + P$ is an M_v augmenting path in G , contradiction. Hence, P is an alternating path from u to v , let $C = uv + P$, note C is an odd cycle because P has even length (by $v = z$ is M_v exposed).

– $\delta(C) \cap M_u = \emptyset$. Since the path is alternating, the only vertex in C not incident to a M_u edge in C is u , but since u is M_u -exposed, $\delta(u) \cap M_u = \emptyset$.

– $M_u \setminus C$ is a maximum cardinality matching in $G \setminus C$. Suppose not, then there is a larger matching M' in $G \setminus C$. And consider $M' \cup \{M_u \cap E(C)\}$, it is a matching in G because $\delta(C) \cap M_u = \emptyset$. And it is larger matching in G than M_u because

$$|M_u| = |M_u \cap E(G \setminus C)| + |M_u \cap E(C)| < |M'| + |M_u \cap E(C)| = |M' \cup \{M_u \cap E(C)\}|$$

contradiction. Hence, C is inessential in G' .

– Hence, $M_u \setminus C$ is a maximum cardinality matching in G/C without including C , so C is inessential in G/C . Since there are $\frac{|C|-1}{2}$ many M_u vertices in C , we know

$$\nu(G) = |M_u| = |M_u \setminus C| + |M_u \cap E(C)| = \nu(G/C) + \frac{|C|-1}{2}$$

so C is a tight odd cycle containing uv , as required.

Lemma 3.12

Let M be a matching, $A \subseteq V$ such that $|M| = \frac{1}{2}(|V| - oc(G \setminus A) + |A|)$. Then all vertices in A are essential.

Proof. Let $v \in A$. Let $A' = A \setminus \{v\}$, $V' = V \setminus \{v\}$, $G' = G \setminus \{v\}$. Since the components of $G \setminus A$ are the same as the components of $G' \setminus A'$, we know

$$\begin{aligned} oc(G \setminus A) &= oc(G' \setminus A') \\ \nu(G') &\leq \frac{1}{2}(|V'| - oc(G' \setminus A') + |A'|) \\ &= \frac{1}{2}(|V| - 1 - oc(G \setminus A) + |A| - 1) \\ &= |M| - 1 \end{aligned}$$

so v is essential. □

Then answer our question, $C \in A'$ reaches a contradiction because C is inessential. Hence, such C exists, and $C \notin A'$. □

3.5 Matching 5

We say an M -alternating tree T is frustrated if $\forall uv \in E, u \in B(T)$, we have $v \in A(T)$.

Proposition 3.13

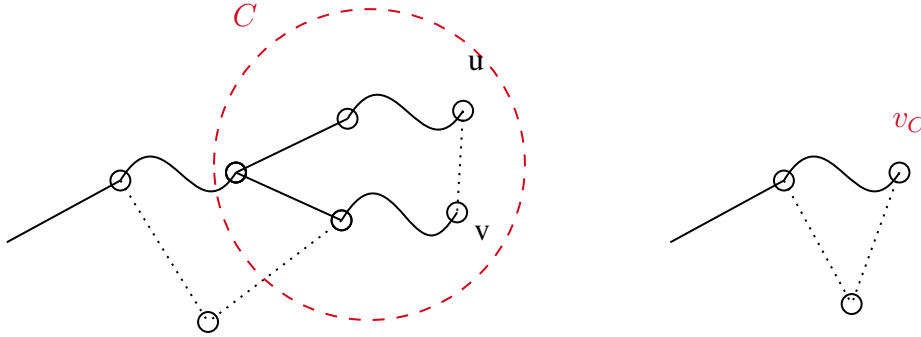
If T is frustrated, then G has no perfect matching.

Proof. Since all neighbors of vertices in $B(T)$ are in $A(T)$, we know $G \setminus A(T)$ has at least $|B(T)|$ many odd components, because each vertex in $B(T)$ in $G \setminus A(T)$ is an odd component. Hence,

$$|oc(G \setminus A(T))| \geq |B(T)| > |A(T)|$$

then by Tutte's Matching Theorem, we know G has no perfect matching. $\square$

Let $u, v \in B(T)$ such that $uv \in E$, then $T + uv$ has a unique odd cycle C (called Blossom). Shrink the Blossom and let $G' = G/C$.



Note:

- Edges in $M \setminus E(C)$ form a matching M' in G' .
- Shrunk Tree T' is M' -alternating in G' .
- Pseudonode v_C is in the set $B(T')$ for the tree T' .

Note: One may need to shrink multiple times.

We say the graph obtained after shrinking (sequentially) Blossoms is a derived graph. $S(v)$ will represent the set of vertices that have been shrunk into $v \in V(G')$, then

$$\forall v \in V(G'), S(v) = \begin{cases} v, & \text{if } v \in V(G) \\ \cup_{w \in C} S(w), & \text{if } v = v_C, \text{ for some Blossom } C \end{cases}$$

Note: $|S(v)|$ is odd, $\forall v \in V(G')$ by definition $|S(v)| = 1$ or it's a sum of odd many odd numbers.

Proposition 3.14

Let G' be a derived graph from G , M' a matching of G' , T' an M' -alternating frustrated tree of G' with all pseudonode in $B(T')$, then G has no perfect matching.

Proof. If G has a perfect matching M , then for any Blossom C , G/C also has a perfect matching $M \setminus C$, hence, G' will have a perfect matching, but G' has an M' -alternating tree, contradiction. $\square$

Proposition 3.15

Let G' be derived graph from G , M' an matching of G' , T' an M' -alternating tree, $uv \in E(G')$ with $u, v \in B(T')$, C' unique cycle (Blossom) in $T' + uv$.
Then $M'' = M' \setminus E(C')$ is a matching for $G'' = G'/C'$ and $T'' = (V(T') \setminus V(C') \cup \{v_{C'}\}, E(T') \setminus E(C'))$ is an M'' -alternating tree in G'' with $v_{C'} \in B(T'')$.

Algorithm 7 Blossom Algorithm for Perfect Matching

Input graph G and matching M of G
Set $M' = M$, $G' = G$
Choose an M' -exposed node r of G' and put $T = (\{r\}, \emptyset)$
while there exists $vw \in E'$ with $v \in B(T)$, $w \notin A(T)$ **do**
 if $w \notin V(T)$, w is M' -exposed **then**
 Use vw to augment M'
 Extend M' to a matching M of G
 Replace M' by M and G' by G
 if there is no M' -exposed node in G' **then**
 Return the perfect matching M' and stop
 else
 Replace T by $(\{r\}, \emptyset)$ where r is M' -exposed.
 end if
 else if $w \in V(T)$, w is M' -covered **then**
 Use vw to extend T
 else if $w \in B(T)$ **then**
 Use vw to shrink and update M' and T
 end if
end while
return G' , M' , T and stop; G has no perfect matching.

Theorem 3.16

Blossom algorithm does $O(n)$ augmentation, $O(n^2)$ shrinks, $O(n^2)$ tree extensions and correctly determines if G has perfect matchings.

Proof. Each augmentation increase $|M'|$ by 1, implies $O(n)$ augmentation. Between two augmentation steps, shrink reduces size of G' by at least 2 vertices implies $O(n)$ shrinks, so total $O(n^2)$ shrinks. Similar for tree extensions. $\square$

Algorithm 8 Blossom Algorithm for Maximum Cardinality Matching

Input graph G and matching M of G
Set $M' = M$, $G' = G$, $\mathcal{T} = \emptyset$
(★) Choose an M' -exposed node r of G' and put $T = (\{r\}, \emptyset)$
while there exists $vw \in E'$ with $v \in B(T)$, $w \notin A(T)$ **do**
 if $w \notin V(T)$, w is M' -exposed **then**
 Use vw to augment M'
 Extend M' to a matching M of G
 Replace M' by M and G' by G
 if there is no M' -exposed node in G' **then**
 Return the perfect matching M' and stop
 else
 Replace T by $(\{r\}, \emptyset)$ where r is M' -exposed.
 end if
 else if $w \notin V(T)$, w is M' -covered **then**
 Use vw to extend T
 else if $w \in B(T)$ **then**
 Use vw to shrink and update M' and T
 end if
end while
 $\mathcal{T} \leftarrow \mathcal{T} \cup \{T\}$; $G' \leftarrow G' \setminus V(T)$; $M' \leftarrow M' \setminus E(T)$
if There exists an M' -exposed node in G' **then**
 go back to (★)
else
 return $M = \cup_{T \in \mathcal{T}} M_T$
end if

Proof. Let $T_1, \dots, T_k$ be the trees in $\mathcal{T}$; M be the final matching. For each T_i , there exists only one M -exposed vertex in T_i because each T_i is an M_{T_i} -alternating tree, so the only M -exposed vertex in T_i is its root, so there are k M -exposed vertices in total. Let $A = \cup_{i=1}^k A(T_i)$. Each vertex in $B(T_i)$ is an odd component of $G \setminus A$ because each T_i is frustrated, all neighbors of vertices in $B(T_i)$ are in A . Hence,

$$oc(G \setminus A) \geq \sum_{i=1}^k |B(T_i)| \geq \sum_{i=1}^k (|A(T_i)| + 1) = |A| + k$$

which implies

$$|M| = \frac{|V| - k}{2} \geq \frac{1}{2}(|V| - oc(G \setminus A) + |A|)$$

so M is a maximum cardinality matching. □

3.6 Matching 6

Definition 3.17: Gallai-Edmonds Decomposition

Let $G = (V, E)$, B be the set of inessential vertices, $C := \{v \in V \setminus B : v \in N_G(B)\}$, $D := V \setminus (C \cup B)$. (B, C, D) is called the Gallai-Edmonds partition/decomposition of G .

Proposition 3.18

Let $T_i, i = 1, \dots, k$ be the frustrated trees found in Blossom algorithm. Then

$$C = \cup_{i=1}^k A(T_i), \quad B = \cup_{i=1}^k (\cup_{v \in B(T_i)} S(v)), \quad D = V \setminus (B \cup C)$$

Note.

- This implies all components of $G[B]$ are odd and C is a minimizer of Tutte-Berge Formula.
- This also implies that Gallai-Edmonds decomposition can be computed in polytime.
- Implies $G[D]$ only has even components. (every vertex in D is M -covered, and it's not matched to A nor B).

Proof. We saw all vertices in $\cup_{i=1}^k A(T_i)$ are essential (by the proof of correctness of the Blossom Algorithm, we know A is the minimizer of Tutte-Berge Formula hence all vertices in it is essential). For all $v \in \cup_{i=1}^k (\cup_{v \in B(T_i)} S(v))$, there exists an even M -alternating path from an M -exposed vertex u to it. Pick such path P , and then $M' = M \triangle E(P)$ is a matching with $|M'| = |M|$, and v is M' -exposed which implies that v is inessential.

- Consider $v \in V \setminus \underbrace{(\cup_{i=1}^k A(T_i) \cup (\cup_{i=1}^k (\cup_{v \in B(T_i)} S(v))))}_{D'}$, and consider $G' = G \setminus v$. Since D' only has even components, we know $oc(G' \setminus C) = oc(G \setminus C \setminus v) > oc(G \setminus C)$, not D is not connected to B , so we removing v will not increas the number of components in B , but only D . Hence

$$\nu(G') \leq \frac{1}{2} (|V| - 1 - oc(G' \setminus C) + |C|) < \frac{1}{2} (|V| - oc(G \setminus C) + |C|) = \nu(G)$$

- Hence, v is essential.

Note.

- $v \in D'$ is not adjacent to a vertex in B , otherwise, if v is M -covered, we can extend M , if it's M -exposed, we can augment M .
- $v \in C$ is adjacent to a vertex in B by the definition of the alternating trees.

□

4 Weighted Matching

4.1 Weighted Matching 1

Minimum Weight Perfect Matching

Given $G = (V, E)$, $c_e \in \mathbb{R}$, $\forall e \in E$, find a perfect matching M of G minimizing $c(M) = \sum_{e \in M} c_e$.
Idea:

$$\begin{aligned} \min \quad & \sum_{e \in E} c_e x_e \\ \text{s.t.} \quad & x(\delta(v)) = 1, \forall v \in V \\ & x \geq 0, x \in \mathbb{Z}^E \end{aligned}$$

and we can have the relaxation as

$$\begin{aligned} (P_M) : \min \quad & \sum_{e \in E} c_e x_e \\ \text{s.t.} \quad & x(\delta(v)) = 1, \forall v \in V, \\ & x \geq 0 \end{aligned}$$

$$\begin{aligned} (D_M) : \max \quad & \sum_{v \in V} y_v \\ \text{s.t.} \quad & y_u + y_v \leq c_{uv}, \forall uv \in E \end{aligned}$$

Note: $Z_{P_M} :=$ optimal value of (P_M) , so $Z_{P_M} \leq c(M)$, $\forall$ perfect matching M . (Notice that every perfect matching's indicator vector is a feasible solution for P_M).

Q: Can we solve our problem by solving (P_M) ?



Every perfect matching has at least one edge with cost 1. However, the optimal value of P_M is 0 because we can give 0.5 to those edges of the triangles.

Theorem 4.1: Birkhoff

Let $G = (V, E)$ be bipartite, $c \in \mathbb{R}^R$, then G has a perfect matching if and only if (P_M) is feasible. Moreover, if (P_M) is feasible, then let M^* be a minimum cost perfect matching, then we have $Z_{P_M} = c(M^*)$.

Proof.

G has a perfect matching $\iff P_M$ is feasible (SKIPPED)

Remaining statement: Algorithmic Proof.

Construct a matching H that corresponds to a optimal solution to (P_M) using Complementary Slackness:

- Let $\bar{y}$ be feasible for (D_M) .
- Let $E^= := \{uv \in E : \bar{y}_u = \bar{y}_v = c_{uv}\}$
- If $G^= := (V, E^=)$ has a perfect matching M , then $x^M, \bar{y}$ satisfy Complementary Slackness conditions, so we are done, we know M is a minimum weighted perfect matching.
- Else, update $\bar{y}$.

But how should we update $\bar{y}$?

Recall at the end of the algorithm for perfect matching on $G^=$, we will be in one of the two situations

- Found a perfect matching M , and it's the min weighted perfect matching in G .
- It finds a frustrated tree in $G^=$.

Idea: Update $\bar{y}'_v$ to get $E_{new}^=$ such that

- $\bar{y}$ is still feasible for (D_M) .
- Current $M \subseteq E_{new}^=$.
- Current $E(T) \subseteq E_{new}^=$.
- At least one edge $uv \in E \setminus E_{old}^= : u \in B(T), v \in V(T)$ is in $E_{new}^=$.

Let $\epsilon = \min\{c_{uv} - \bar{y}_u - \bar{y}_v : u \in B(T), v \notin V(T)\}$, and let $\bar{y}_u^* = \begin{cases} \bar{y}_u + \epsilon, & \forall u \in B(T) \\ \bar{y}_u - \epsilon, & \forall u \in A(T) \\ \bar{y}_u, & \forall u \notin V(T) \end{cases}$.

- $\bar{y}^*$ is still feasible for (P_M) . Since the graph is bipartite, no $uv \in E$ such that $u, v \in B(T)$. If $u \in B(T), v \in A(T)$, then $\bar{y}_u^* + \bar{y}_v^* = \bar{y}_u + \bar{y}_v$. If $u \in A(T), v \notin V(T)$, then $\bar{y}_u^* + \bar{y}_v^* = \bar{y}_u + \bar{y}_v - \epsilon \leq \bar{y}_u + \bar{y}_v \leq c_{uv}$. If $u \in B(T), v \notin V(T)$, then

$$\bar{y}_u^* + \bar{y}_v^* = \bar{y}_u + \bar{y}_v + \epsilon \leq \bar{y}_u + \bar{y}_v + c_{uv} - (\bar{y}_u + \bar{y}_v) = c_{uv}$$

If $u, v \notin V(T)$, then $\bar{y}_u^* + \bar{y}_v^* = \bar{y}_u + \bar{y}_v$.

- $M \subseteq E_{new}^=$. Consider any edge uv in $E(T)$, then it has one end in $A(T)$ and the other end in $B(T)$, so we know $\bar{y}_u^* + \bar{y}_v^* = \bar{y}_u + \bar{y}_v = c_{uv}$, so $uv \in E_{new}^=$. That is, $M \subseteq E(T) \subseteq E_{new}^=$.

□

Algorithm 9 Min Weight Perfect Matching Algorithm for Bipartite Graphs

Let y be a feasible solution to (P_M) , M a matching of $G^=$
 If M is a perfect matching of G , return M and stop
 Set $T \leftarrow (\{r\}, \emptyset)$ where r is an M -exposed node of G
while not stopped **do**
 while there exists $vw \in E^=$ with $v \in B(T)$, $w \notin V(T)$ **do**
 if w is M -exposed **then**
 Use vw to augment M
 if there is no M -exposed node in G **then**
 Return the perfect matching and stop
 else
 Replace T by $(\{r\}, \emptyset)$ where r is M -exposed
 end if
 else
 Use vw to extend T
 end if
end while
if every $vw \in E$ with $v \in B(T)$ has $w \in A(T)$ **then**
 Stop, G has no perfect matching
else
 Let $\epsilon = \min\{c_{vw} - y_v - y_w : v \in B(T), w \notin V(T)\}$
 Replace y_v by $y_v + \epsilon$ for $v \in B(T)$, $y_v - \epsilon$ for $v \in A(T)$.
end if
end while

Note.

- $M \subseteq E^=$ all the time, so if we find a perfect matching, it will be a min weight one.
- Stopping points: either we find a perfect matching in G or we find a frustrated tree in G (not $G^=$), so there is no perfect matching.
- The loop can only run polynomially many times because for every iteration, one more edge will be added to T , so the algorithm will terminate in polynomial time.